BEACHES

Who pays for beach closures

When bacteria counts spike, the beach closes. The bill for testing, signs and lost summer days lands in places most swimmers never see.

By **Owen Brandt**
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On a hot weekend, a closed beach looks like a simple call: the water tested badly, so the sign goes up. Behind the sign is a chain of costs that runs from county health departments to city park budgets to the businesses that live on summer traffic.

Few of those costs show up on a receipt. Nobody bills the family that drove two hours and turned around in the parking lot, or the lifeguard sent home at noon. But someone pays for every step, from the bottle of lake water to the advisory on the website. Following that money also explains something that puzzles swimmers every summer: why one beach closes three times in July while another, across the same lake, almost never closes at all.

The first bill: testing

Testing is the first cost and the most predictable. Samples are collected on a schedule, rushed to a lab and read the next day, which is why many closures trail the conditions that caused them.

The standard method is a culture test. A technician mixes the sample with a nutrient that E. coli bacteria feed on, incubates it for 18 to 24 hours and counts what grew. Most Great Lakes states post an advisory when the count tops 235 colonies per 100 milliliters of water. By the time anyone knows that number, the people who were swimming when the bottle was filled have gone home, and the lake may already have cleared.

Much of the money for that work starts in Washington. The federal BEACH Act, passed in 2000, lets the Environmental Protection Agency send grants to coastal and Great Lakes states to monitor swimming beaches and warn the public. The program runs about $10 million a year nationwide, divided by a formula that weighs each state's swimming season, its miles of beach and how many people use them. Budget writers in Washington have tried to zero it out more than once. Congress has kept it alive.

The states report back on how they spend it. Minnesota's most recent report shows about $200,000 for its Lake Superior beach program, most of it for sampling crews, lab fees and the advisory website. Wisconsin's share is somewhat larger and has to cover far more beaches on two Great Lakes.

The federal money is a floor, not a ceiling. Cities, counties and park districts add their own, and the gap between what the grant covers and what a beach needs is where the testing schedule gets set.

How often is often enough

Here is the arithmetic that never makes it onto the beach sign. A beach sampled once a week gets one look at its water every seven days. A beach sampled every morning gets seven. Bacteria counts often spike for a day or two after a heavy rain and then fall, so a once-a-week beach can go a whole summer without an advisory, not because its water is cleaner but because nobody sampled it on the wrong day.

Same water, two schedules

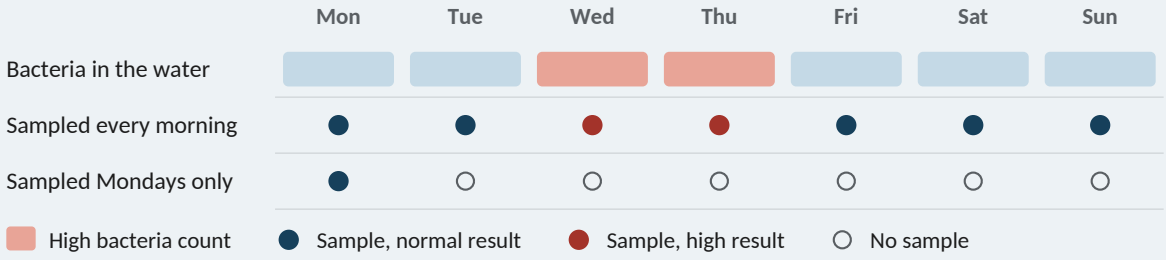


Illustration. After a Tuesday-night storm, bacteria counts stay high for two days. A beach sampled every morning catches both days. A beach sampled only on Mondays never sees them.

Beach program staff in Minnesota, Wisconsin and Illinois made the same point in interviews: the number of advisories a beach posts says as much about how often it is tested as about the water.

Testing schedules follow the money. Every sample means a drive to the beach, a cooler and a lab fee. Federal rules ask states to rank their beaches by risk and use and to test the busiest most often. In Door County, Wisconsin, where the county samples dozens of public beaches on both the bay side and the Lake Michigan side of the peninsula, the most crowded beaches are tested more often in midsummer and the quiet ones less.

The county's numbers took some untangling. Its own beach log and the state's annual summary disagreed on how many samples were taken at two beaches last August. County staff explained the gap: after a high reading, crews go back the next day, and again if they need to, until the water tests clean. The county log counted those return trips. The state summary, built for the federal report, counted only scheduled samples. Both figures were accurate. Only the county's showed what it spent chasing a single bad result.

The faster test

There is a faster option, and Chicago pays for it. The Chicago Park District moved its lakefront beaches to a rapid DNA test, known as qPCR, several summers ago. A sample taken early in the morning can produce a result by early afternoon, while swimmers are still in the water.

The rapid test does not count live E. coli. It measures the DNA of enterococci, another group of bacteria that signal fecal contamination, using a method the EPA has approved for beach monitoring. It costs more per sample than a culture, often two to three times as much once staff time is counted, and it needs a lab instrument that costs tens of thousands of dollars.

How the two tests compare

	Culture test	Rapid DNA test (qPCR)
What it counts	Live E. coli that grow from the sample	DNA from enterococci bacteria
Time to a result	18 to 24 hours	About 3 to 4 hours
Cost per sample	Lower	Often two to three times higher, plus a lab instrument
Advisory line	235 colonies per 100 mL in most Great Lakes states	A DNA threshold set under EPA criteria
In this story	Park Point beaches; Door County beaches	Chicago lakefront beaches

Sources: EPA beach monitoring guidance; state beach program reports; interviews with beach program staff.

That raises the question at the center of beach testing: who pays for the faster answer, and who decides whether it is worth the money? Under the federal program, each state picks its methods, but the state rarely owns the beach. A city or park district that wants same-day results generally pays the difference itself, and on a lakefront with more than 20 public beaches, the difference adds up. In Chicago, that cost sits in the park district's own budget.

The lost Saturday

The costs that are hardest to see fall on the people who rent out the paddleboards.

At Park Point in Duluth, the long sand spit that shelters the harbor from Lake Superior, the owner of a small snack and rental stand near the beach keeps a notebook of advisory days next to the cash box. Last July, two of them fell on Saturdays.

On a good July Saturday, the stand takes in about $1,800 from paddleboard rentals, ice cream and cold drinks. On the first advisory Saturday last summer, it took in $240.

"Nobody sends me a bill," the owner said. "I just don't make the money."

No budget has a line for that loss. The federal grant pays for the sample, and the city pays for the sign. The stand absorbs the rest. What the owner wanted was faster news. Because the culture test takes a day, the advisory that emptied the beach that Saturday was based on water collected Friday morning. The next sample, taken Saturday, came back clean on Sunday, and the sign came down that afternoon.

“I’ve been doing this 11 summers, and you’re the first person who’s asked what it costs me,” the owner said.

The bill for fixing it

The harder costs come later. Cities that want fewer closures end up paying for storm-sewer work, shoreline repairs and gull and goose control, and that money rarely comes out of the same budget that pays for the tests.

Testing tells a beach manager the water is dirty. It does not say why. Finding the source is a separate job with a separate bill. Gulls are a common culprit at Great Lakes beaches. So are storm sewers that empty near the swimming area after a heavy rain, failing septic systems upstream and wet sand that holds bacteria at the water’s edge.

Door County has spent years on that second job. At several of its beaches, crews regraded the sand, moved stormwater away from the swimming area and planted rain gardens to soak up runoff before it reached the lake. The work was paid for with a mix of state and federal grants and county money, and at the beaches where it was done, advisories dropped.

That kind of fix costs far more than a summer of samples, and it usually comes out of public works budgets, not health budgets. The agency that pays for the testing is seldom the one that can fix what the testing finds. The business that loses a Saturday has no budget at all.

Who decides

For swimmers, the result is a patchwork. A beach tested every day with a same-day method will post more advisories, and more timely ones, than a beach tested weekly by culture. On a state map, the first beach looks dirtier. It may well be the safer place to swim.

Federal rules set the floor: states must monitor their highest-priority beaches and tell the public when the water fails. Everything else, from how often to test to which method to use and who pays for it, is left to each state and to the cities and counties that own the sand. With the federal grant roughly flat for years, those decisions are made one beach budget at a time.

Back at Park Point, the stand owner has a simpler wish. “Test it more often, and tell us sooner,” the owner said. “I’d chip in for that.”

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Great Lakes Dispatch is a nonprofit newsroom based in Chicago that covers the environment, economy and communities of the Great Lakes region. Read more at greatlakesdispatch.org.